

Facial immersion in cold water enhances cerebral blood velocity during breath-hold exercise in humans

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Kjeld T, Pott FC, Secher NH. Facial immersion in cold water enhances cerebral blood velocity during breath-hold exercise in humans. *J Appl Physiol* 106: 1243–1248, 2009. First published January 29, 2009; doi:10.1152/jappphysiol.90370.2008.—The diving response is initiated by apnea and facial immersion in cold water and includes, besides bradycardia, peripheral vasoconstriction, while cerebral perfusion may be enhanced. This study evaluated whether facial immersion in 10°C water has an independent influence on cerebral perfusion evaluated as the middle cerebral artery mean flow velocity (MCA V_{mean}) during exercise in nine male subjects. At rest, a breath hold of maximum duration increased the arterial carbon dioxide tension (PaCO_2) from 4.2 to 6.7 kPa and MCA V_{mean} from 37 to 103 cm/s (mean; $\sim 178\%$; $P < 0.001$). Similarly, during 100-W exercise, a breath hold increased PaCO_2 from 5.9 to 8.2 kPa ($P < 0.001$) and MCA V_{mean} from 55 to 113 cm/s ($\sim 105\%$), and facial immersion further increased MCA V_{mean} to 122 cm/s ($\sim 88\%$; both $P < 0.001$). MCA V_{mean} also increased during 180-W exercise (from 47 to 53 cm/s), and this increment became larger with facial immersion (76 cm/s, $\sim 62\%$; $P < 0.001$), although PaCO_2 did not significantly change. These results indicate that a breath hold diverts blood toward the brain with a $>100\%$ increase in MCA V_{mean} , largely because PaCO_2 increases, but the increase in MCA V_{mean} becomes larger when combined with facial immersion in cold water independent of PaCO_2 .

diving response; heart rate; mean arterial pressure; arterial carbon dioxide tension

LIKE OTHER ANIMALS, HUMANS possess a diving response that is initiated during apnea and augmented with facial immersion in cold water (3, 9, 18). The diving response includes peripheral vasoconstriction, a reduced cardiac output (CO), and bradycardia, and also the exercise-induced tachycardia is attenuated (12). The cardiovascular response elicited by the diving response may divert blood toward the brain, as demonstrated in the rat (15).

Trained divers demonstrate a strong diving response, reduced postapnea acidosis and oxidative stress, and a reduced sensitivity to hypoxia and hypercapnia (9, 18). At the end of a dry breath hold, trained divers demonstrate reduced muscle oxygenation and a large increase in middle cerebral artery mean flow velocity (MCA V_{mean}) (16), providing evidence for the oxygen-conserving effect of the diving response. Several factors may affect cerebral blood flow during diving. Cerebral blood flow and the transcranial Doppler determined MCA V_{mean} increase during exercise (10), suggesting an increase in blood flow to a large part of the brain, independent of the concomitant increase in mean arterial pressure (MAP) (11). Also, the exercise-induced increase in MCA V_{mean} is modified

by the arterial carbon dioxide tension (PaCO_2) (13, 14), and hypoxia enhances blood flow, especially to phylogenetically older regions of the brain (4).

This study evaluated whether MCA V_{mean} increases when trained divers are exposed to facial immersion in 10°C water, exercise, and intermittent apnea. Because apnea, facial immersion, and exercise may each have an independent influence on MCA V_{mean} , the effect of facial immersion was separately evaluated and when combined with a breath hold. Maximum duration apnea at rest and 30-s apnea during exercise were chosen for this protocol, to represent the circumstances for the underwater rugby player during warm-up and a game, respectively. Two intensities of exercise were chosen, with 100 W representing the game, while 180 W was chosen to represent the circumstances during the warm-up and endurance training. We hypothesized that MCA V_{mean} would increase during apnea as PaCO_2 increases, but that facial immersion in cold water would carry an independent influence on MCA V_{mean} .

METHODS

The nine subjects who participated in the study were healthy nonsmoking men at 28 yr of age (21–34 yr) (median with range) with a weight of 75 kg (68–92 kg) and a height of 178 cm (172–192 cm). Every subject provided written and informed consent to the protocol, as approved by the Ethics Committee of Copenhagen (KF 01 271889). During the main study, the subjects were sitting upright on the cycle ergometer provided with a water tub to immerse their face by bending the neck $\sim 45^\circ$. All subjects were fasting for 2 h before any participation in the study and, on a day before the study, maximal exercise was carried out on a cycle ergometer (Monark 829E, Varberg, Sweden) at 55–60 rpm, and the workload was increased by 30 W every minute until exhaustion [330–390 W, which calculates to that the subjects possessed a maximal oxygen uptake of $58 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ ($49\text{--}62 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) (2)].

The proximal segment of the left MCA was insonated (Multidop X, DWL, Sipplingen, Germany) through the posterior temporal ultrasound window. Once the optimal signal-to-noise ratio was obtained, the probe was covered with adhesive ultrasonic gel (Tensive, Parker Laboratories, Orange, NJ) and secured with a headband. The V_{mean} of MCA correlates to a ^{133}Xe clearance-determined cerebral blood flow (11), and it demonstrates a marked response to PaCO_2 (10). MAP, cardiac stroke volume (SV), heart rate (HR), and thus CO were monitored noninvasively with the finger volume clamp method from the index finger of the dominant hand (6). SV, HR, and CO were calculated from the blood pressure waveform using the Modelflow algorithm, incorporating age, sex, height, and body weight (6). Arterial blood samples were obtained at rest, with 5-min intervals during exercise, and at the end of each apnea through a catheter (1.1 mm, 20 gauge) in the brachial artery of the nondominant arm with connection

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to a transducer for continuous flow of saline (Baxter, Uden, The Netherlands). Blood-gas analyses were performed immediately after sampling, using an automated self-calibrating blood-gas machine (ABL 725, Radiometer, Copenhagen, Denmark) and evaluated for pH, bicarbonate, lactate, P_{aCO_2} , and the arterial oxygen tension (P_{aO_2}).

All breath holds were timed by the investigator with countdown, and the subjects were instructed to inhale submaximally. On the first day of the main study, the subjects performed three 30-s breath holds to maximize breath-hold time and the diving response (18). Then, after 1 min, the subjects performed three 30-s breath holds separated by 30 s of recovery. On a separate day, the subjects were instructed to perform three 30-s breath holds as detailed, and, after 1 min, the subjects performed three breath holds of maximum duration separated by 30 s of recovery. Following further 5 min of recovery, the subjects cycled at 100 W on the ergometer for 12 min with 30-s breath holds separated by 30 s of free breathing. To evaluate whether MCA V_{mean} was enhanced by facial immersion in cold water during moderate exercise, the subjects performed 15 min of 180-W cycling exercise with normal breathing after 5 min of recovery. On a third day, this protocol was repeated with facial immersion in a 10°C water bath while breathing through a snorkel, without using a nose clip.

Data were sampled at 200 Hz on a Windaq Lite Data Acquisition DI-194 system (Akron, OH). Obvious artifacts were removed by inspection from the spectral outline of the Doppler signal (MATLAB 7.1, Mathworks), and beat-to-beat values were calculated (BeatScope 1.1a; TNO).

Variables are presented as means $\pm$ SD for baseline, as the average of three breath holds of maximum duration at rest, the average of the intermittent breath holds during 100-W cycling, and for 15 min of 180-W cycling, with and without facial immersion in cold water. Three-second intervals were chosen to represent the beginning and the end of each breathing and breath-hold period. Friedman's repeated-measures analysis of variance on ranks was used to determine whether significant changes manifested over time or between circumstances, and such changes were located with Student-Newman-Keul's test. A P value <0.05 was considered statistically significant.

RESULTS

Both at rest and during exercise, MCA V_{mean} demonstrated a peak shortly after the onset of each breath hold, with and without facial immersion followed by a sharp drop (Fig. 1, Table 1). Then MCA V_{mean} increased to reach a maximum at the end of the breath hold. HR, SV, and thus CO and also MAP decreased at the onset of the breath hold, and HR continued to

decrease, while SV, CO, and MAP stabilized or recovered somewhat at the end.

At rest, a 30-s breath hold left MCA V_{mean} statistically unchanged, with and without facial immersion. However, during a breath hold of maximum duration (178 ± 19 s), MCA V_{mean} increased from 37 ± 23 to 103 ± 15 cm/s ($\sim 178\%$; $P < 0.001$; Fig. 2), and P_{aO_2} decreased after each breath hold ($P < 0.001$; Table 2). P_{aCO_2} did not significantly change during the first breath hold, but increased from 4.1 ± 1.1 kPa before the second and third breath hold, and pH decreased after the second and third breath hold ($P < 0.001$), while lactate and bicarbonate did not significantly change. During the breath hold of maximum duration, HR decreased from 82 ± 7 to 63 ± 7 beats/min ($P < 0.05$), and CO decreased from 8.5 ± 2.7 to 2.7 ± 2.1 l/min, while MAP and SV did not significantly change.

100-W exercise. During 100-W cycling exercise with intermittent breath holds, the MCA V_{mean} response was consistent from the third breath hold with an increase from 55 ± 11 to 113 ± 20 cm/s ($\sim 105\%$; $P < 0.001$; Fig. 3, Table 1). During the breath hold, HR dropped (from 126 ± 26 to 88 ± 28 beats/min; $P < 0.05$), while MAP, SV, and CO did not significantly change. P_{aO_2} decreased after each breath hold ($P < 0.001$; Table 2). P_{aCO_2} was higher after the last breath hold compared with resting values ($P < 0.001$). pH, lactate, and bicarbonate did not significantly change.

During 100-W exercise and facial immersion in cold water with intermittent breath holds, MAP increased from 80 ± 19 to 99 ± 29 mmHg and HR fell from 117 ± 11 to 78 ± 13 beats/min ($P < 0.05$; Fig. 3, Table 1). SV and CO did not significantly change but decreased to a minimum shortly after the beginning of each breath hold and then rose toward the end. Facial immersion and intermittent breath holds increased MCA V_{mean} from 65 ± 12 to 122 ± 27 cm/s ($P < 0.05$).

A breath hold increased MCA V_{mean} from 55 ± 11 to 65 ± 12 cm/s, and facial immersion increased MCA V_{mean} further (68 ± 13 to 76 ± 13 cm/s; $P < 0.05$, Table 1). Facial immersion decreased HR immediately before (117 ± 11 vs. 126 ± 14 beats/min) and after initiating the breath hold (109 ± 16 vs. 120 ± 22 beats/min), and HR further

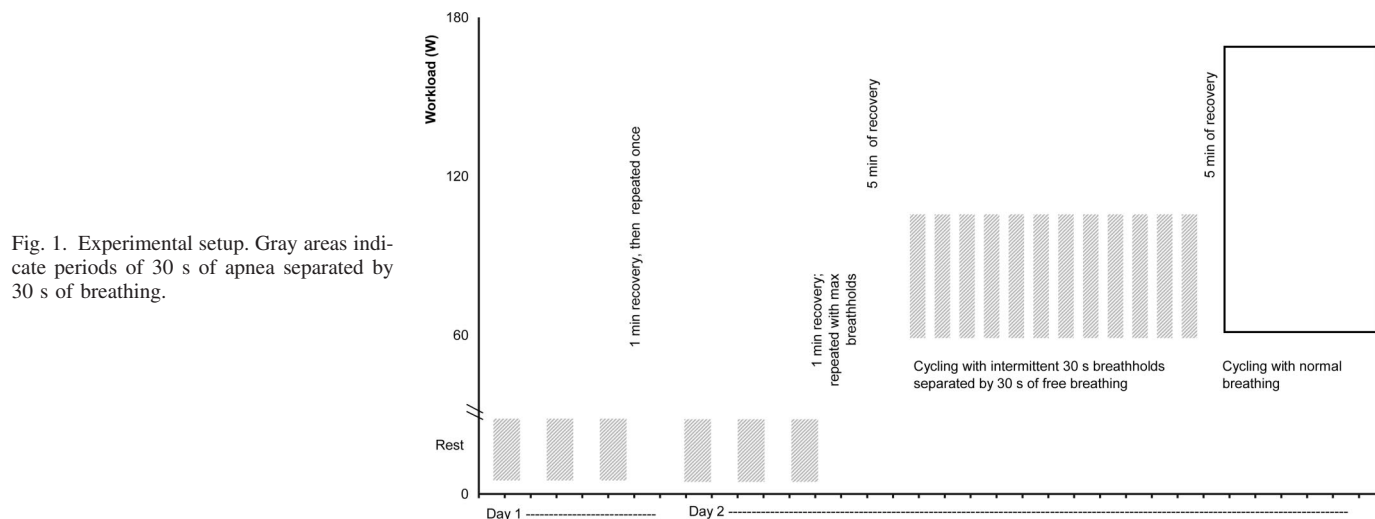


Fig. 1. Experimental setup. Gray areas indicate periods of 30 s of apnea separated by 30 s of breathing.

Table 1. MCA V_{mean} , MAP, HR, SV, and CO during 10 cycles of 30-s breathing followed by 30-s apnea while cycling at 100 W, with and without facial immersion in 10°C water

	Breathing		Breath Hold	
	Start	End	Start	End
MCA V_{mean} , cm/s				
Rest				
Control	35 ± 8	35 ± 10	34 ± 11	37 ± 11
Facial immersion	45 ± 10	39 ± 10	34 ± 9	43 ± 15
Exercise				
Control	116 ± 20	68 ± 13*†	55 ± 11*†	113 ± 21†
Facial immersion	127 ± 40	78 ± 13†	65 ± 12†	122 ± 27†
MAP, mmHg				
Rest				
Control	97 ± 11	89 ± 11	81 ± 17	80 ± 18
Facial immersion	99 ± 14	88 ± 19	85 ± 19	86 ± 21
Exercise				
Control	102 ± 16	82 ± 14	79 ± 11	87 ± 10
Facial immersion	97 ± 29	78 ± 20	78 ± 19	81 ± 23
HR, beats/min				
Rest				
Control	68 ± 7	76 ± 10	60 ± 8	55 ± 5
Facial immersion	68 ± 2	76 ± 5	70 ± 13	59 ± 11
Exercise				
Control	102 ± 24	126 ± 14	120 ± 22*	88 ± 28*†
Facial immersion	90 ± 15	117 ± 11*	109 ± 16*	78 ± 13*
SV, ml				
Rest				
Control	61 ± 15	55 ± 3	44 ± 8	44 ± 6
Facial immersion	79 ± 30	67 ± 24	53 ± 16	63 ± 29
Exercise				
Control	125 ± 46	106 ± 32	103 ± 25	113 ± 32
Facial immersion	145 ± 36	122 ± 30	119 ± 32	128 ± 38
CO, l/min				
Rest				
Control	4.1 ± 1.4	4.1 ± 0.5	2.7 ± 0.8	2.4 ± 0.5
Facial immersion	5.3 ± 1.9	5.1 ± 1.7	3.7 ± 1.3	3.6 ± 1.6
Exercise				
Control	12.6 ± 6.2	13.4 ± 3.5	12.0 ± 2.5	10.1 ± 3.2
Facial immersion	12.7 ± 2.1	14.1 ± 3.0	12.7 ± 3.0	10.3 ± 2.5

Values are means ± SD; $n = 7$ subjects ($n = 8$ subjects for MCA V_{mean}). MCA, middle cerebral artery; V_{mean} , mean flow velocity; MAP, mean arterial pressure; HR, heart rate; SV, stroke volume; CO, cardiac output. *Different from facial immersion ($P < 0.05$). †Different from previous phase ($P < 0.05$).

decreased at the end of a breath hold (78 ± 13 vs. 88 ± 28 beats/min; $P < 0.05$).

180-W exercise. HR increased from 89 ± 19 to 155 ± 9 beats/min ($P < 0.001$), while MAP and SV did not significantly change (Fig. 4). CO increased from 4.6 ± 1.7 to 10.6 ± 2.5 l/min ($P < 0.001$), and MCA V_{mean} from 47 ± 7 to 53 ± 11 cm/s. PaO_2 , PaCO_2 , pH, lactate, and bicarbonate did not significantly change (Table 2).

When exercise at 180 W was combined with facial immersion in cold water, HR increased from 89 ± 15 to 153 ± 15 beats/min ($P < 0.001$), while MAP and SV did not significantly change (Fig. 5). Exercise with facial immersion in cold water increased CO from 5.1 ± 1.9 to 12.5 ± 2.9 l/min and MCA V_{mean} from 54 ± 9 to 76 ± 15 cm/s ($P < 0.05$, Table 2).

DISCUSSION

The main finding of this study is that MCA V_{mean} increases markedly during a breath hold, both at rest and during exercise,

since PaCO_2 increases. However, facial immersion influenced MCA V_{mean} independently from PaCO_2 in trained divers.

Rapid fluctuations in MAP may induce the initial increase in MCA V_{mean} with each breath hold, and the following sharp drop in MAP, SV, CO, and MCA V_{mean} could reflect that deep inspiration preceded the breath hold followed by a Valsalva-like maneuver (17). Decreases in HR during each breath hold,

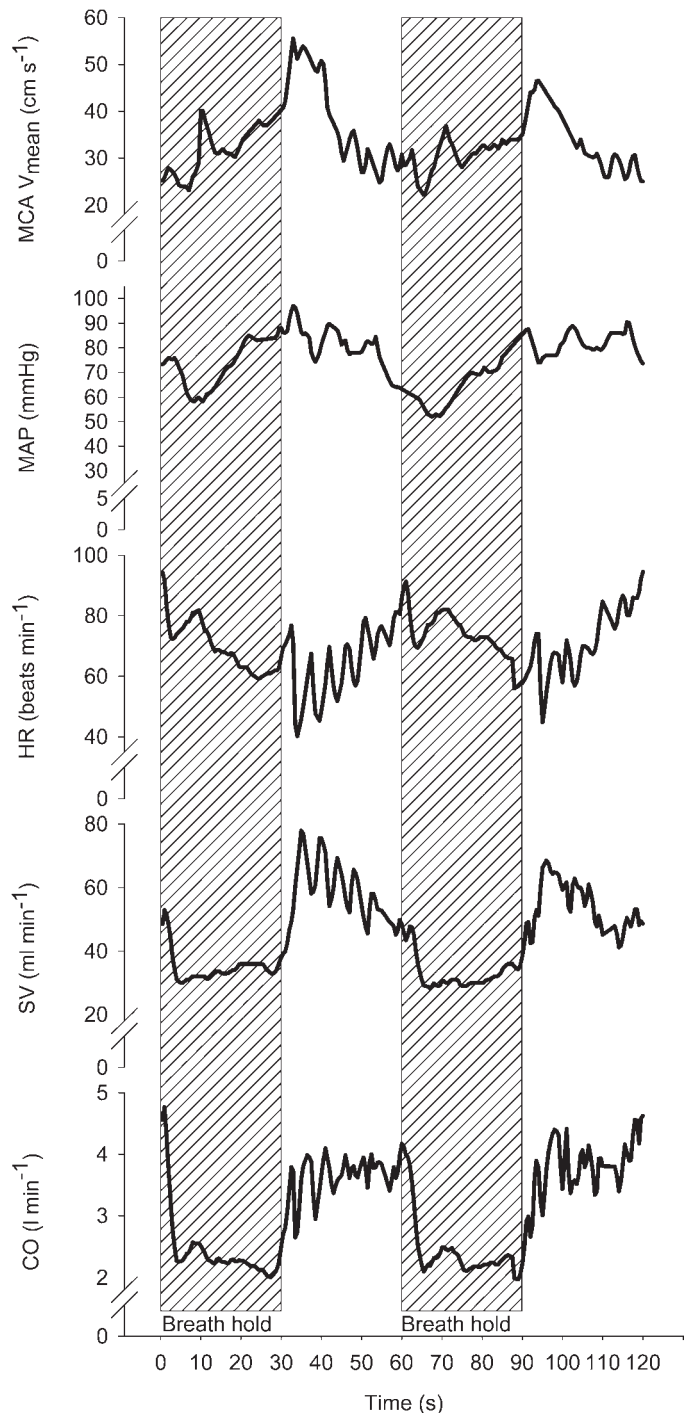


Fig. 2. Middle cerebral artery (MCA) mean flow velocity (V_{mean}), mean arterial pressure (MAP), heart rate (HR), stroke volume (SV), and cardiac output (CO) for 30 s of breathing followed by 30 s of apnea at rest with facial immersion in 10°C water for one subject.

Table 2. Blood-gas variables at rest and during a breath hold with values for both rest and 100- and 180-W exercise

	Breathhold No.	Time, min	pH		PaO ₂ , kPa		PaCO ₂ , kPa	
			Breathing	Breath hold	Breathing	Breath hold	Breathing	Breath hold
Rest (maximum breath hold)	1		7.41±0.05	7.39±0.03	11.8±0.67	7.7±0.9*	5.3±1.0	6.2±0.3
	2		7.50±0.07*	7.38±0.07*	15.2±2.6	8.0±1.3*	4.1±1.1	6.6±0.6*
	3		7.48±0.02*	7.36±0.03*	15.3±2.3	7.6±1.7*	4.2±1.0	6.7±0.6*
100 W (30-s breath hold)	1		7.44±0.08	7.40±0.06	11.8±0.7	7.7±0.9*	5.9±0.7	7.2±0.7
	3		7.42±0.07	7.39±0.09	12.8±0.1	10.1±1.4	6.2±1.6	7.1±1.3
	4			7.37±0.05		9.7±3.6		6.9±2.7
	6		7.39±0.05		10.0±1.9		6.2±0.2	
	7			7.35±0.07		7.5±2.3		7.5±2.3
	9		7.38±0.09		9.2±2.3		7.7±1.4	None
	10			7.35±0.07		9.4±3.6		7.7±2.4
180 W (normal breathing)	12		7.40±0.05	7.35±0.06	11.8±1.4	7.3±1.5*	6.0±0.7	8.2±0.6*
		0	7.38±0.08		12.0±3.6		5.5±0.8	
		5	7.35±0.07		10.8±2.3		5.8±0.5	
		10	7.35±0.02		11.0±2.5		5.7±0.6	
		15	7.36±0.02		11.5±3.0		5.6±0.6	
		Recovery 5 min	7.39±0.04		13.6±5.1		5.4±0.6	

Values are means ± SD. PaO₂, arterial oxygen tension; PaCO₂, arterial carbon dioxide tension. *Different from resting values ($P < 0.001$).

along with the even lower values reached during facial immersion, indicate that the diving response was elicited. Furthermore, apnea during dynamic exercise tended to reduce CO and increase MAP, which may be related to the concomitant reduction in HR supporting that the diving response was elicited (5). Sterba and Lundgren (20) found that the diving response wears off after reaching a plateau from the third cycle of apnea. In the present study, the diving response seemed unaffected by repeated periods of apnea, but subjects performed a 30-s apnea during 100-W exercise, and they felt it was close to a maximum performance. Brown et al. (8) did not find any significant changes in CO during cold face stimulation with normal breathing at rest in nondivers, and with only a small increase in MCA V_{mean} . MAP, SV, and CO all fell in the first 25 s of the breath hold and then rose at the end. Of note, these variables also increased with a combination of exercise and facial immersion in cold water.

Using Finapres, we obtained information on MAP and SV (7), assuming that aortic impedance remains stable (22). However, the present study focused on changes rather than absolute values. Transcranial Doppler monitors blood velocity rather than volume flow, and changes in the diameter of the insonated vessel could modulate velocity independently of flow (1). With magnetic resonance imaging, the MCA diameter remains stable in healthy individuals during hypocapnia (21), hypercapnia, and lower body negative pressure (19), suggesting that the MCA is not involved in regulation of cerebral vascular resistance. However, even small changes in its diameter could influence blood velocity, and the spatial resolution of magnetic resonance of ~0.5 mm is not sufficient to exclude that changes in vascular diameter may affect MCA V_{mean} . The assumption of a stable MCA diameter may thus be invalid during the circumstances of the present study with dynamic changes in blood pressure, intrathoracic pressure, and PaCO₂. The rise in MAP could increase transmural pressure and distend the MCA, leading to an underestimate of the actual cerebral blood flow increase, especially at the onset of exercise with a breath hold.

Cerebral blood flow and MCA V_{mean} increase during exercise (10), independent of the concomitant increase in MAP (11), but the increase is modified by PaCO₂ (13, 14). PaCO₂

increases during a breath hold, and the increase in MCA V_{mean} was related closely to PaCO₂. Yet the increase in MCA V_{mean} was further augmented during facial immersion in cold water with a breath hold during exercise and independent of a breath hold during 180-W exercise.

Nondivers can reduce their PaO₂ to 8.0 kPa (normal range 11–14 kPa) and increase PaCO₂ to 6.0 kPa (normal range 4.3–6.4 kPa); yet trained breath-hold divers will usually endure a breath hold until PaO₂ has fallen to 4.7 kPa and PaCO₂ has increased to 6.7 kPa (9). In the present study, the subjects endured a decrease in PaO₂ to 7.6 kPa and an increase in PaCO₂ to 6.7 kPa at rest. PaO₂ decreased to 7.3 kPa, and PaCO₂ increased to 8.2 kPa during exercise at 100 W combined with a breath hold. Accordingly, we confirm the tolerance to an increase in PaCO₂ in trained divers, but not the decrease in PaO₂ (9).

Exercise increases lactate and, in turn, hydrogen ion concentration in blood, thereby increasing cerebral blood flow (14). Since the diving response induces peripheral vasoconstriction, it could promote anaerobic metabolism in working muscles, and an augmented H⁺ could be assumed. Individuals accustomed to (scuba) diving react to facial immersion in water with a pronounced reduction in HR and a rise in MAP (18). Sport divers with 7–10 yr of experience develop less postapnea acidosis and oxidative stress, but probably reduced sensitivity to progressive hypoxia and hypercapnia (9, 18). This might explain why Brown et al. (8) found only a small increase in brain blood flow at rest and during facial cooling in nondivers, and we found significant changes in pH only during apnea at rest.

These results suggest that a breath hold diverts blood toward the brain with more than a 100% increase in MCA V_{mean} , according to the increase in PaCO₂. However, the increase in MCA V_{mean} was larger by facial immersion in cold water independent of PaCO₂.

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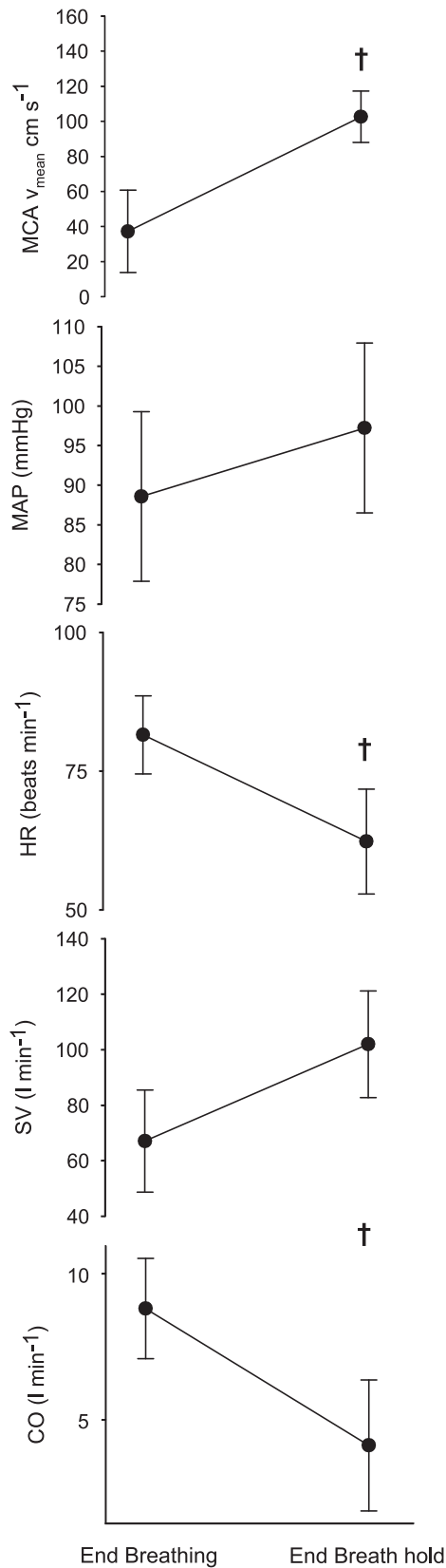


Fig. 3. MCA V_{mean} , MAP, HR, SV, and CO for 30-s breathing followed by maximum duration apnea at rest. Values are means $\pm$ SD for 6 subjects (MCA V_{mean} for 7 subjects). †Different from rest ($P < 0.05$).

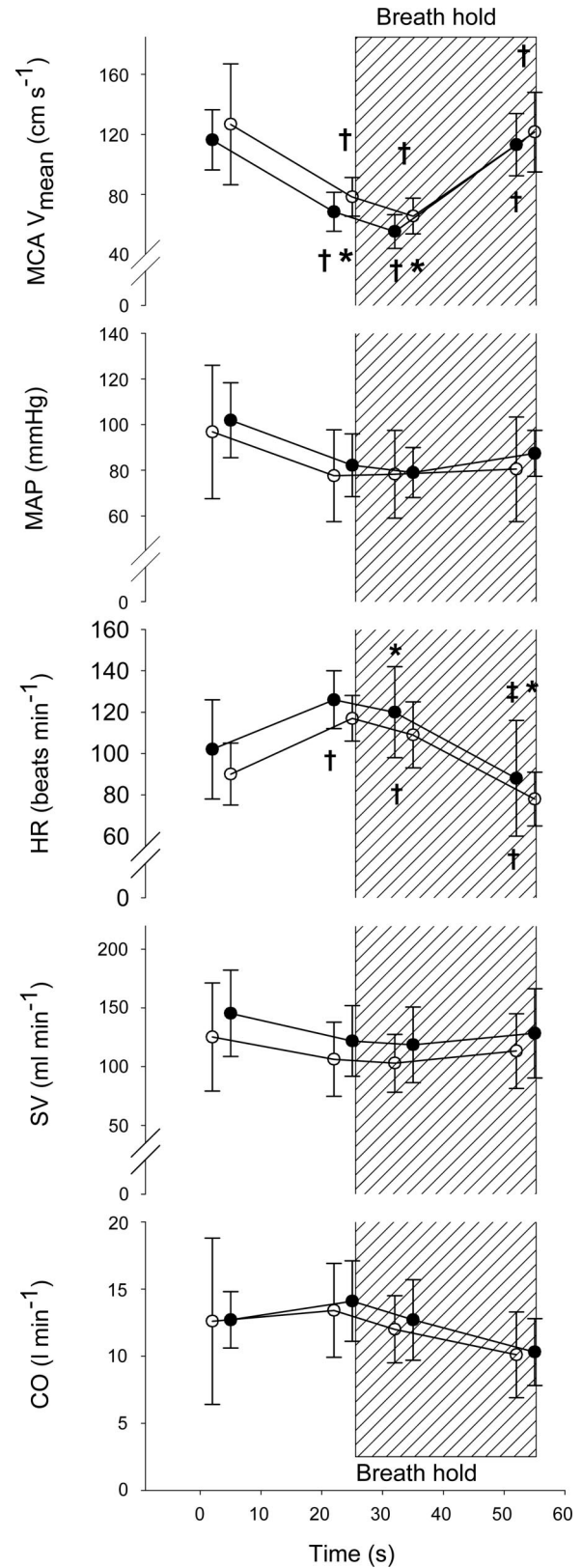


Fig. 4. MCA V_{mean} , MAP, HR, SV, and CO during 10 cycles of 30-s breathing followed by 30-s apnea, while cycling at 100 W with (●) and without facial immersion (○) in 10°C water. Values are means $\pm$ SD for 6 subjects (MCA V_{mean} for 8 subjects). *Different from facial immersion ($P < 0.05$). †Different from previous phase ($P < 0.05$).

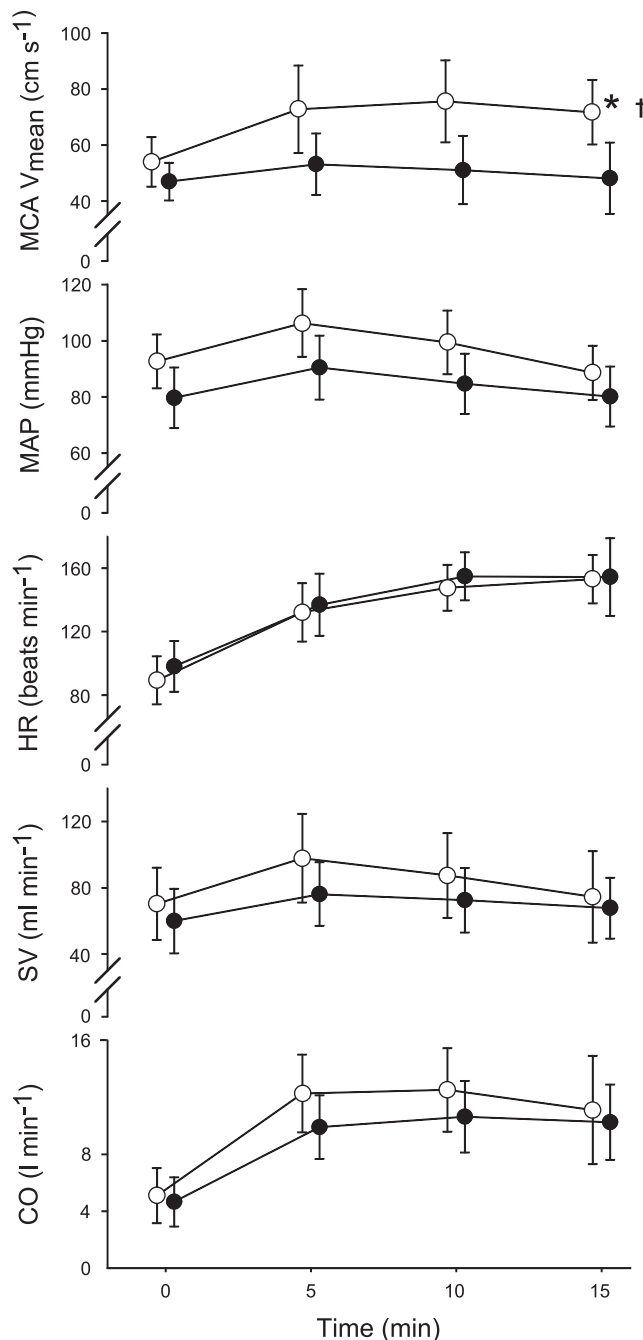


Fig. 5. MCA V_{mean} , MAP, HR, SV, and CO during 15-min 180-W cycling exercise with (○) and without facial immersion (●) in 10°C water. Values are means $\pm$ SD for 7 subjects (MCA V_{mean} for 8 subjects). *Different from facial immersion ($P < 0.05$). †Different from previous phase ($P < 0.05$).

GRANTS

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